

Geometric Constructions Using Triangle Congruence

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Geometric construction is the process of drawing figures using only a compass and a straightedge.

Copying and Bisecting a Segment

A **segment** can be copied or bisected accurately using geometric construction.

Copying preserves the exact length; bisecting divides the segment into two equal halves.

Copying and Bisecting an Angle

An **angle** can be copied or bisected using a compass and straightedge.

Copying preserves the exact measure; bisecting divides the angle into two equal parts.

Constructing Triangles

A **triangle** can be constructed when sufficient measurements are given, such as **SSS**, **SAS**, or **ASA**.

These are the same conditions you learned for triangle congruence.

Why the Construction Is Unique

The **triangle congruence postulates and theorems** justify why a constructed triangle is unique.

If two triangles satisfy the same SSS, SAS, or ASA conditions, they must be congruent — so only one triangle shape is possible.

Secondary Parts of a Triangle

The **perpendicular bisectors**, **angle bisectors**, **medians**, and **altitudes** are called the **secondary parts** of a triangle.

Each type of secondary part intersects at its own special point.

Triangle Centers

Circumcenter

The **circumcenter** is the intersection of the perpendicular bisectors of a triangle's sides.

Incenter

The **incenter** is the intersection of a triangle's angle bisectors.

Centroid

The **centroid** is the intersection of a triangle's medians.

Orthocenter

The **orthocenter** is the intersection of a triangle's altitudes.

Triangle Centers at a Glance

Circumcenter — perpendicular bisectors

Incenter — angle bisectors

Centroid — medians

Orthocenter — altitudes

Examples

Example 1

Construct the perpendicular bisector of $\overline{AB} = 12$ cm.

Answer: The perpendicular bisector splits $\overline{AB}$ into two equal segments:

$$AM = MB = 6 \text{ cm}$$

Example 2

Construct the bisector of a 70° angle.

Answer: The bisector splits the angle into two equal parts:

$$35^\circ + 35^\circ = 70^\circ$$

Example 3

Construct a triangle with two sides measuring 6 cm and 8 cm and an included angle of 60° .

Answer: This gives exactly one triangle, guaranteed unique by the **SAS Congruence Postulate**.

Practice Exercises

Practice 1

Construct a segment congruent to $\overline{AB} = 8$
cm.

Practice 2

Construct the perpendicular bisector of a segment measuring 10 cm. Determine the length of each resulting segment.

Practice 3

Construct an angle congruent to $\angle XYZ = 58^\circ$.

Practice 4

Construct the bisector of a 96° angle. Find the measure of each resulting angle.

Practice 5

Construct a triangle with side lengths 5 cm, 6 cm, and 8 cm. Identify the congruence postulate that guarantees a unique triangle.

Practice 6

Construct a triangle with two sides measuring 7 cm and 9 cm and the included angle measuring 55° .

Activities

Activity 1

Construct a segment congruent to $\overline{CD} = 7$
cm.

Activity 2

Construct the perpendicular bisector of a segment measuring 14 cm. Determine the length of each resulting segment.

Activity 3

Construct an angle congruent to $\angle ABC = 72^\circ$.

Activity 4

Construct the bisector of a 124° angle. Find the measure of each resulting angle.

Activity 5

Construct a triangle with side lengths 4 cm, 7 cm, and 9 cm. Identify the congruence postulate that guarantees a unique triangle.

Activity 6

Construct a triangle with two sides measuring 8 cm and 10 cm and the included angle measuring 68° .

Thank you!